

The ESL Programming Language

Brian G. Lucas

Table of Contents

Chapter 1. Introduction	5
Chapter 2. Syntactic Elements	6
2.1 Comments	6
2.2 Reserved Words	6
2.3 Identifiers	6
2.3.1 Identifier Namespaces	6
2.4 Numeric Literals	7
2.5 Character Literals	7
2.6 String Literals	7
Chapter 3. Types and Declarations	8
3.1 Integer Type	8
3.2 Enumerated Type	9
3.3 Fixed Point Type	9
3.3.1 Fixed point literals	10
3.4 Floating Point Type	10
3.4.1 Floating point literals	10
3.5 Data Reference Type	10
3.6 Procedure Reference Type	10
3.7 Array Type	10
3.7.1 Fixed Sized Arrays	11
3.7.2 Unknown Sized Arrays	11
3.8 Record Type	11
3.9 Field Attributes	12
3.9.1 At	12
3.10 Type Attributes	12
3.10.1 Size	12
3.10.2 Alignment	13
3.10.3 Bit Order	13
3.10.4 Memory Order	13
3.10.5 Rounding Mode	13
3.10.6 Access Restrictions	14
3.10.7 Input and Output	14
3.11 Built-in Types	14
Chapter 4. Variables	15
4.1 Variable Declarations	15
4.2 Variable Attributes	15
4.2.1 Global	15
4.2.2 Weak	15
4.2.3 External	16
4.2.4 Segment	16
Chapter 5. Expressions	16
5.1 Expression Boundaries	16
5.1.1 Type Inference	16
5.1.2 Changing Types in Expression	17
5.1.2.1 Type Casting	17

5.1.2.2 Type Conversion	18
5.2 Fundamental Terms	18
5.2.1 Type Queries	19
5.3 Unary Operations	20
5.4 Binary Operations	20
5.4.1 Multiplicative Operations	20
5.4.2 Additive Operations	21
5.4.3 Comparison Operations	21
5.4.4 Boolean And	21
5.4.5 Boolean Or	21
5.5 Intrinsic Operations	21
5.5.1 Absolute Value	21
5.5.2 Maximum Value	22
5.5.3 Minimum Value	22
5.5.4 Square Root	22
5.5.5 Count Leading Zeros	22
5.5.6 Count Trailing Zeros	22
5.5.7 Population Count	22
5.5.8 Rotate right and rotate left	23
5.5.9 Bit Reverse	23
5.5.10 Unpack Byte Array	23
5.5.11 Pack Byte Array	23
5.5.12 Get Low or High Half	23
5.5.13 Create Operand from Two Halves	23
5.5.14 Set	24
5.5.15 Zero	24
5.5.16 Find Non-Zero Length	24
5.5.17 New Allocation	24
5.5.18 Delete Allocation	24
Chapter 6. Statements	25
6.1 Declarative Statements	25
6.1.1 Type Statement	25
6.1.2 Const Statement	25
6.1.3 Variable Statement	25
6.1.4 Alias Statement	25
6.2 Assignment Statement	26
6.2.1 Scalar Assignment	26
6.2.2 Assignment Operators	27
6.2.3 Record Copy	27
6.2.4 Array Copy	27
6.3 Control Statements	28
6.3.1 If Statement	28
6.3.2 Exit Statement	28
6.3.3 Loop Statement	28
6.3.4 While/Do Statement	29
6.3.5 Do/While Statement	29
6.3.6 For Statement	29

6.3.7 Return Statement	29
6.3.8 Assert Statement	29
6.4 Formatting Statements	30
6.5 Asm Statement	31
Chapter 7. Procedures	31
7.1 Normal Procedures	32
7.2 Methods	32
7.3 Procedure Attributes	33
7.3.1 Inline/Noinline	33
7.3.2 Global	33
7.3.3 Weak	33
7.3.4 External	33
7.3.5 Segment	34
7.4 Parameter Attributes	34
7.5 Forward Procedures	34
7.6 Procedure References	34
Chapter 8. Packages	35
8.1 Package Continuation	35
Chapter 9. Programs and Scope	36
9.1 Import Statement	36
9.2 Conditional Compilation	36
9.3 Compilation Style	36
Chapter 10. Custom Formatting	37
Chapter 11. Language Syntax Summary	39
11.1 Notation	39
11.2 Program Syntax	39
11.3 Package Syntax	39
11.4 Procedure Syntax	39
11.5 Declaration Statement Syntax	40
11.6 Executable Statement Syntax	40
11.7 Expression Syntax Summary	41
Chapter 12. Library Support	41
Chapter 13. ESL For C Programmers	41
13.1 Preprocessor	41
13.1.1 Include	42
13.1.2 If and ifdef	42
13.2 Declarations	42
13.3 Typedefs	42
13.4 Enumerations	42
13.5 Pointers and Arrays	42
13.6 Pointers to Procedures	42
13.7 Parameters and Arrays	43
13.8 Statements	43
13.8.1 Assignment Statements	43
13.8.2 Assignment Within an Expression	44
13.8.3 If Statements	44
13.8.4 Switch Statements	44

Chapter 14. Building and Using the ESL Compiler	45
14.1 Prerequisites	45
14.2 Building the ESL compiler	45
14.2.1 Getting the source	45
14.2.2 Installing LLVM	46
14.2.3 Building the compiler	46
14.3 ESL Command Line Options	46
14.3.1 -D[asf]	46
14.3.2 -mtarget	47
14.3.3 -Idir	47
14.3.4 -ofile	47
14.3.5 -Ooopt	47
14.3.6 -Aaopt	47
14.3.7 -Ffopt	48
14.3.8 -M	48
14.3.9 -g	48
14.4 Compilation Flow	48

1 Introduction

ESL is a programming language designed to be used for efficient programming of embedded and other "small" systems. ESL an acronym for Embedded Systems Language (pronounced: "ESS-el").

ESL is a typed compiled language with features that allow the programmer to dictate the concrete representation of data values. This distinguishes it from languages which implement only "abstract" types or types whose representation is architecture-dependent. The programmer can dictate the details of data representation, including such things as "endian-ness" and the exact placement of bits, which are necessary in dealing with external representations of data layout, e.g., communication protocols or device registers.

ESL is not really a “new” programming language - all the elements have probably been seen in other programming languages. In many respects, it is a conventional language whose features, for the most part, will be familiar to programmers who have had experience with any of the compiled languages introduced in the past 50 years.

If the ESL syntax bears some resemblance to Google's programming language *Go*, that is mainly due to common inspiration. The syntax for ESL was mostly in place before *Go* was made public. There are two exceptions. One is the syntax for methods, this was shamelessly stolen from *Go*. The second is not yet implemented.

2 Syntactic Elements

2.1 Comments

There are two ways to comment text: by line or by block. Line comments start with “//” and end at the end of the line. Block comments start with “/*” and end with “*/”. Block comments do not nest. Block comments may contain line comments.

2.2 Reserved Words

There are no reserved words in ESL. There are many keywords that have special meaning only in specific contexts. These keywords can also be used as ordinary identifiers. Keywords are always lower case.

2.3 Identifiers

The first character of an identifier must be from the set {A-Z,a-z}, remaining characters can be chosen from the set {A-Z,a-z,0-9,_}. The system reserves identifiers starting with a “_” for predefined symbols so they will not restrict programmer defined symbols. In addition, the identifier consisting of a single “_” has a special meaning. It is not entered into the symbol table and can be used to specify anonymous fields in records, placeholders in enumeration, or ignored returned values from a call..

Identifiers may also contain UTF8 sequences any place an alphabetic character is valid. (Identifiers with UTF8 sequences may not be handled correctly in the LLVM backend, by assemblers, or by linkers.)

Declared Identifiers may be type names, variable names, procedure names, package names, field names, enumeration constant names, or aliases.

2.3.1 Identifier Namespaces

There are two visibility classes of identifiers: public and private. The public identifiers are nested at four levels: universal, global, package, and procedure. Universal identifiers are pre-defined by the compiler. Global identifiers declared at the outermost nesting level of a program. Package identifiers are declared within a package. Packages may be nested. Finally, procedure identifiers are declared within a procedure. There is no further nesting within a procedure, i.e., no “blocks”, and procedures can not be nested.

Private identifiers include enumeration constants, fields within a record, and methods. They are accessed by prefixing the enumeration type name, the record variable name, or a variable of the method type.

2.4 Numeric Literals

Numbers are by default decimal. For other number bases, a two-character prefix is used. The first character is always a zero, the second character, always lower case, indicates the number base:

0b binary - Only digits **0-1** are allowed.

0o octal - Only digits **0-7** are allowed.

0x hexadecimal - Only digits **0-9** and letters **A-F** or **a-f** are allowed.

In any number, after the first digit (which follows the optional prefix), an underline “**_**” is allowed for readability.

Examples:

```
399                // decimal
0b0010_0001        // binary
0o02_56             // octal
0x01_F4             // hexadecimal
1_999_999           // decimal
```

2.5 Character Literals

Character literals are unsigned integer constants. Character literals are enclosed within single quotes, the literal is either any single ASCII character (excluding the single quote and newline) or an escape sequence. Escape sequences start with a backslash and include:

**** - represents a single backslash

\n - represents a newline

\r - represents a carriage return

\f - represents a form feed

\t - represents a horizontal tab

\b - represents a backspace

\v - represents a vertical tab

\xXX - represents an 8-bit character with value given by the two hex digits **XX**

\uXXXX - represents a 16-bit (e.g. unicode) character with value given by four hex digits

\UXXXXXXXX - represents a 32-bit (e.g. unicode) character with value given by eight hex digits

2.6 String Literals

A string literal is a constant array of bytes optionally terminated by a NUL. Each byte is either an ASCII character, an 8-bit escape as defined in the above character literal section, or a part of a UTF8

sequence. The character escapes which generate greater than 8-bit values are converted into a sequence of bytes by UTF8 encoding. String literals bounded by double quotes are NUL terminated, those bounded by single quotes are not. Some examples:

```
const string1 = "This is nul terminated";
const string2 = 'This is not nul terminated';
const string3 = "The following is unicode greek delta \u0394.";
```

String literals may be defined in sequential pieces. If any of the pieces, except for the final one, are NUL terminated, the NUL will be removed. Examples:

```
const string4 = "This is " "a single NUL"
                "terminated string";
```

3 Types and Declarations

Types in ESL are not abstract in the sense that they are designed to be concrete descriptions of containers for a set of values. When instances of types are allocated, the assignment to bit positions and memory addresses depends on architecture-dependent defaults and which can be overridden by the use of optional type attributes.

There are two units of allocation:

- *bits* - the atomic unit of data
- *bytes* - the smallest sequence of bits that has a memory address, in all target architectures

Types consist of the integral types, floating point types, references (pointers), and aggregate types. Integral types are unsigned or signed and are always sub-ranges. Enumerations are a special type of unsigned type. Signed integers are assumed to have twos complement representation. Aggregate types are either arrays or records.

Enumerations and records can be “extended” from base types. Details on how each of these types are extended will be discussed in the section where the specific type is defined.

Types are declared with type statements. Type statements may occur anywhere in the statement flow.

3.1 Integer Type

The integer types are specified by giving their lowest and highest values. The resulting type may be signed or unsigned, depending on the values given. The representation of unsigned sub-ranges always includes zero. The representation of signed sub-ranges is symmetrical (in the twos-complement sense) around zero.

Examples:

```
type percent: 0..100;           // an unsigned type
type srange: -1000..1000;       // a signed type
```


3.2 Enumerated Type

Enumerated types are a mapping onto the unsigned integers. They are defined by listing identifiers associated with a value. For a given value there may be one or more identifiers associated with that value. The identifiers will be assigned the next numeric value by the compiler, or optionally, can be given a specific numeric values by the programmer.

An enumerated type consists of a contiguous range of values from zero to some maximum value, there are no gaps. All of the values need not be represented by identifiers.

Operations on enumerated types are limited to assignment and comparison. However, casting to unsigned integer types is allowed.

Each enumerated type is distinct. That is, two different enumerated types are not compatible for assignment or comparison.

An enumerated type can be extended by adding additional values.

Examples:

```
type answer: (yes, no);           // yes=0, no=1
type foo: (low=5, medium, high=10) // medium=6,
    // values 0..4, 7..9 are part of the enumeration
type bar: (A, (B,C,D), E, (F,G)=9); // A=0, B=C=D=1, E=2, F=G=9

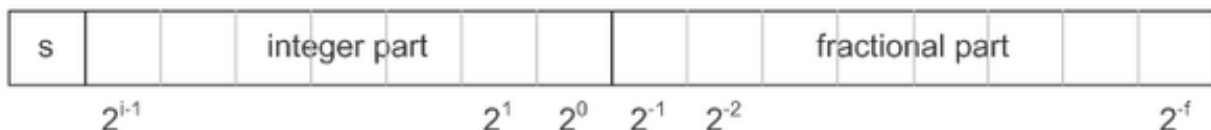
type fuzzy(answer): (maybe, possibly);
    // extends answer: yes=0, no=1, maybe=2, possibly=3
```

The size of an enumeration is the minimal number of bytes necessary to hold the largest defined value. For all the enumerated types in the previous example, the size that would be needed for all three, i.e., answer, foo, and fuzzy, would be one byte in most architectures.

3.3 Fixed Point Type

The fixed point type is only available in the experimental “FixedPoint” branch of the compiler source.

Fixed point types are either unsigned or signed. They are scaled integers with explicitly declared scale. They are declared with total width (in bits) and fraction width (in bits). In signed fixed point types the sign bit is part of the total width. The size of the integer part is the width minus the fractional part minus 1 for the sign, if any.



Examples:

```
type q32_32: _ufix<32,32>; // all 32 bits are fraction
type q16_15: _sfix<16,15>; // 1 bit of sign, 15 bits of fraction
type degrees: _ufix<8, 3>; // 5 bits of integer, 3 bits of fraction
```

3.3.1 Fixed point literals

Fixed point literal constants always have decimal base. They must start with a decimal digit and must have a decimal point.

Examples:

```
const half: _ufix<16,16> = 0.5;
const minusthird: _sfix<16,15> = -0.333333;
const one: _sfix<16,15> = 1.0;
```

In the last example, the constant is not in range of the type, but is allowed and adjusted to be the next smaller number that is in range.

3.4 Floating Point Type

There are two built-in floating point types: `_float32` and `_float64`.

3.4.1 Floating point literals

Fixed point literal constants always have decimal base. They must start with a decimal digit and must have a decimal point. They may optionally have an explicit exponent starting with `e` or `E`.

Examples:

```
const pi: _float64 = 3.14159265358979323846;
const large _float32 = 10.0E+10;
const small _float32 = 10.0e-10;
```

3.5 Data Reference Type

This is just a pointer to another type, the base type.

It is possible that a programmer wants to reference a type that has not yet been declared. This is possible by referencing the type name and then later completing the type definition for that name.

Examples:

```
type byte: 0..255;           // the base type
type ptr_to_byte: @byte;    // a pointer to the base type

type ptr_to_forward: @forward; // forward reference
type forward: sometype;      // the real declaration is later
```

3.6 Procedure Reference Type

A procedure reference type is a pointer to a procedure. See section 7.6

3.7 Array Type

Arrays are aggregates of a base type. The base type is accessed by the indexing operation. Array

indices are unsigned and always start at zero.

Array slicing is an operation that allows access to a contiguous sub-array of values. A slice operator is square brackets enclosing an offset and length separated by a colon.

3.7.1 Fixed Sized Arrays

The index specification of a fixed sized array can be either an expression that is a compile-time constant, or the name of an unsigned type (e.g. an enumeration).

Examples:

```
type array1: [100]_uint;
type enum: (zero, one, two, three);
type array2: [enum]enum;
```

3.7.2 Unknown Sized Arrays

Arrays of indeterminate size are allowed as targets of a reference (pointer).

Examples:

```
type pstring: @[]_byte;
type argv: @[]@[]_byte;           // Unix argv type
```

3.8 Record Type

Record types are a way to define an aggregation of dissimilar types as new type. A record is defined as an order list of *fields*. Each field may be of any type. The following example demonstrates how the type base is declared with two fields: one and two.

```
type base:
{
  one: _uint32;
  two: _uint32;
};
```

The final field of a record may have a type of unknown size:

```
type ident:
{
  length: _uint;
  name:   []_byte;
};
```

As mentioned earlier, a record type is one of the types that can be 'extended'. The following example extends the type 'base' declared above:.

```
type newbase(base):
{
  three: _uint16;
  four:  _uint8;
};
```

The new extended type, ‘**newbase**’, includes all the fields that were declared in the type ‘**base**’, and two additional fields. New fields in the extended type are always added at the end. If the base record has a final field of unknown size, it will not become part of the extended record. This allows that final variable length field in the base to be “overlaid” with fields in the extended record.

3.9 Field Attributes

There is currently only one field attribute.

3.9.1 At

The **at** attribute allows the programmer to specify the offset at which the field should be placed. For packed records, the offset value is in bits, otherwise the offset value is in bytes.

```
type notpacked:
{
  a: _byte;           // offset 0 bytes
  b: _byte: at(3);    // offset 3 bytes – padding placed before
  c: _byte;           // offset 4 bytes
};
type packed:
{
  a: _byte;           // offset 0 bits
  b: _byte: at(8);    // offset 8 bits – padding placed before
  c: _byte;           // offset 9 bits
}: packet, lsb;
```

3.10 Type Attributes

Type attributes can be used to alter the way instances of that type are allocated or behave. Allocation attributes alter the size, alignment, bit-order, and byte-order of the memory structure. Usage attributes alter the way that instances of the type can be used.

When declaring types, any attributes given apply only to the outermost type. In the following:

```
type array: []@_byte: align(16);
```

the alignment attribute applies to the array, not the pointer or base type `_byte`.

3.10.1 Size

A size attribute overrides the default (target dependent) size of a type:

bits(cexpr) - instances of this type should be exactly this many bits in size.

size(cexpr) - similar to **bits**(), except the size is in memory-units.

At most, only one of these attributes should be present. If no alignment attributes are present, the size of a type may be increased to a target-dependent value.

3.10.2 Alignment

Alignment attributes change the way a type is positioned in memory:

packed - when the type is allocated no padding occurs at the bit level nor the memory-unit level.

mempacked - that padding occurs only up to the next memory-unit.

align(*cexpr*) - padding up to an alignment in memory, or a record offset, will be made to the *cexpr* byte boundary.

At most, only one of these attributes should be present. If none are present, then allocation will use whatever the target architecture prefers. On many architectures alignment is type dependent .

3.10.3 Bit Order

The bit order attributes affect allocation within an aggregate, record or array, that is packed.

lsb - allocation begins at the least significant bit within a memory-unit and with the memory-unit with the lowest address, i.e. "little-endian"

msb - allocation begins at most significant bit within a memory-unit and with the memory-unit with the highest address, i.e. "big-endian" also known as "network order"

At most, only one of these attributes should be present. If neither attribute is given, the compiler will use whatever the target architecture prefers.

3.10.4 Memory Order

The memory order attributes affect how a multiple byte type is stored in memory. The compiler will "byte-swap", if necessary when loading or storing to insure that the byte-order is correct for the target machine.

le - types that span multiple bytes will have the least significant byte at the first allocated byte.

be - types that span multiple bytes will have the most significant byte at the first allocated byte.

At most, only one of these attributes should be present. If neither attribute is given, the compiler will use whatever the target architecture prefers.

3.10.5 Rounding Mode

The fixed point types may be declared with a rounding mode attribute. These attributes determine the rounding algorithm at assignment points when least significant bits will be lost. The default is **rnddn**. Generally, using any rounding mode other than **rnddn** will incur run-time overhead and an increase in code size.

Let **R** be the result of rounding and **x** be the value before rounding.

rnddn – round down, towards negative infinity: **R <= x**

This is generally the default for most architectures as it is simple truncation.

rndup – round up, towards positive infinity: **R >= x**

rndtz – round toward zero: **abs(R) <= abs(x)**

rndfz – round away from zero: **abs(R) >= abs(x)**

rndns – round to next smaller: exact half-way values and less are rounded down

rndnl – round to next larger: exact half-way values and greater are rounded up

rndne – round to nearest even: exact half-way values are rounded such that the rounded value has a zero as a low order bit

This is also known as convergent rounding or banker's rounding.

rndno -round to nearest odd: exact half-way values are rounded such that the rounded value has a one as a low order bit

3.10.6 Access Restrictions

These attributes allow the programmer to restrict access to allocations of this type.

ro - read only

wo - write only

At most, only one of these attributes should be present. If neither attribute is given, the default is to allow both read and write.

3.10.7 Input and Output

These attributes indicate to the compiler that allocations of this type has external side effects. These attributes are most often used to describe device registers which do not behave as normal memory.

in - loads of instances of this type must not be optimized away because reading may cause external side effects

out - stores to instances of this type must not be optimized away because writing may cause external side effects

io – indicates both **in** and **out**

Some C-based programming languages use the keyword “volatile” to lump together both attributes **in** and **out**.

3.11 Built-in Types

There are a small number of pre-defined types. These include:

_boolean an enumerated type consisting of the constants **false** and **true**.

_byte an unsigned integral type with a range equal to that which can be stored in the target architecture's minimal addressable unit. For all supported architectures, this is the same as **_uint8**.

_uint8, _uint16, _uint32, _uint64 are unsigned types of 8, 16, 32, and 64 bits respectively.

`_int8`, `_int16`, `_int32`, `_int64` are signed types of 8, 16, 32, 64 bits respectively.

`_uint` and `_int` are unsigned and signed integral types with a range equal to that which can be stored in the target architecture's "natural word size". This is usually the size of an integer register.

`_float32`, `_float64` are floating point types.

`_memory` is a synonym for the `[]_byte` type.

`_address` is a synonym for the `@_memory` type.

Because these type identifiers start with “_” they can not be redefined by the programmer. For convenience, the alias “`boolean`” is predefined to be the same as “`_boolean`”; but can be redefined by the programmer. The original boolean constants are always available by the names `_boolean.false` and `_boolean.true`.

4 Variables

4.1 Variable Declarations

Variable declarations instantiate a type. The visibility and lifetime of the variable depend on the nesting level at which the declaration appears. When declaring a variable with a type which has an array of unknown size, the size must be specified. All variables of static scope are initialized to zero. Scalar variables of procedure scope are initialized to zero if the `-Az` flag is present.

```
type arr1: [8]_byte;
type arr2: []_uint32;
var x1: arr1;           // known size
var x2: arr2(16);       // specify size
```

4.2 Variable Attributes

4.2.1 Global

The global attribute indicates that the variable name has global linkage. The attribute may optionally specify a name by which it will be known globally.

```
var foo1: _int: global;           // global linkage name "foo1"
var foo2: _int: global("F00");    // global linkage name "F00"
```

4.2.2 Weak

The weak attribute is similar to the global attribute except that the linkage type is global but weak.

```
var bar1: _int: weak;             // weak global linkage name "bar1"
var bar2: _int: weak("BAR");      // weak global linkage name "BAR"
```

4.2.3 External

The external attribute indicates that the variable instantiation is external to the program being compiled. The attribute may optionally specify a name (via a string) of the external variable. Another option is to specify an address.

```
var baz1: _int: external;           // external linkage name "baz1"
var baz2: _int: external("BAZ");   // external linkage name "BAZ"
var vect: @[32]_uint: external(0x20000000); // external at a fixed address
```

4.2.4 Segment

The section attribute indicates that the variable should be placed in the named section by the linker. The format of section names is linker-specific.

Example:

```
var foo: _int: section(".far_memory");
```

5 Expressions

5.1 Expression Boundaries

The notion of an “expression boundary” is used by the compiler for type inference and for type conversion. The following is a list of expression boundaries:

1. The start of the evaluation of the right-hand side of an assignment or const statement
2. Evaluation of actual parameter in a procedure call
3. Evaluation of a returned value in a procedure
4. Explicit type casting
5. Explicit type conversion

5.1.1 Type Inference

In many cases, as an expression is being evaluated, the type of an identifier can be inferred based on a “target type”.. The target type is set at expression boundaries.

1. As a simple example of the first case, take the assignment of an enumeration constant to an enumeration variable:

```
type enum: (A, B, C);
var foo: enum;
foo = A;                // same as foo = enum.A
```

Since enumeration constants are in a private namespace associated with the enumeration type, the name “A” would be invisible, it would have to be referred to by its qualified name “enum.A”. However, assignment allows the type of the left-hand side (the “target type”) to guide the type of the right-hand side. Any names on the right hand side which are unknown are interpreted in the context of the type.

The target type is also set when evaluating an expression for an actual parameter or returned value. Given the above definition of “enum”, and example of the second case is:

```
proc bar(a: enum);
proc baz()
{   bar(B);           // same as bar(enum.B)
}
```

If the target type has an aggregate type, type inference controls the construction of aggregate expressions. For example:

```
type R: { a: boolean; b: _uint; c: _int; };
var r: R;
r = { true, 1, -1 };
```

5.1.2 Changing Types in Expression

Inevitably, one will have to convert the type of a variable into one of more preferable type for a particular situation. There are two mechanisms for doing this within an expression, casting and conversion.

5.1.2.1 Type Casting

Casting converts a value of one type to another type without changing the bits that represent the value except:

1. the value may be sign extended to the left, or
2. the value may be truncated from the left.

Thus, casting may be thought as “bit pattern preserving”. Casting often does not involve any run-time overhead.

Probably the most common casting situation is changing a value from one bit width size to another. Widening of scalars occurs automatically, as there is no loss of information. [Currently, this may not work everywhere, e.g. formal to actual parameters.] Narrowing, on the other hand, may require the programmer to use explicit type casting. In the following example, the variable u32, a 32 bit variable, is assigned to the variable u16 which has a size of 16 bits.. The following example shows how the variable is properly cast using the name of the destination type. In this case, the name _uint16 which is that of the variable two.

```
var u32:    _uint32;
var u16:    _uint16;

u16 = _uint16(u32);
```

It is often necessary to convert pointer types. This, too, is done by casting:

```

type R:
{
    one:    _uint32;
    two:    _uint32;
};

type pR: @R;
var vR: pR;

vR = pR(mem.Alloc(R?size));

```

One thing to remember is because the underlying bit pattern remains essentially unchanged, the following cast may be surprising:

```

var u32: _uint32;
var f32: _float32;

u32 = _uint32(f32);

```

The resulting value of u32 is the bit pattern that represents a floating point value.

5.1.2.2 Type Conversion

Type conversion changes a value from one type to another type and may change the bit layout of the value. Type conversion may be thought of as “value preserving”. The syntax for type conversion is similar to casting, but with a “*” following the type name.

```

var u32: _uint32;
var f32: _float32;

u32 = _uint32*(f32);

```

In this case the resulting value of u32 is an integer with a value that is the integer part of the floating point value.

5.2 Fundamental Terms

Precedence as presented in the following table.

Operator	Description	Associativity
.	Package dereference	left-to-right
. . [] @ ?	Record dereference Method introduction Brackets (array subscript) Pointer dereference Property inquiry	left-to-right
()	Parentheses (function call)	left-to-right

Operator	Description	Associativity
_abs _max _min _sqrt _clz _clznz _ctz _ctznz _pop _ror _rol _bitrev _unpkbe _unpkle _lo _hi _splice	Absolute value Maximum value Minimum value Square root Count leading zeros Count trailing zeros Population count Rotate right or left Bit reverse Unpack byte array to scalar Get low or high half of scalar Build scalar from high and low half	
+ - ~ !	Unary plus (no-operation) Unary minus Bitwise negation Boolean not	right-to-left
* / % << >> &	Multiplication Division Modulus Bitwise shift left Bitwise shift right Bitwise AND	left-to-right
+ - ^	Addition Subtraction Bitwise OR Bitwise XOR	left-to-right
== != < <= > >=	Relational equal to Relational not equal to Relational less than Relational less than or equal to Relational greater than Relational greater than or equal to	left-to-right
&&	Boolean AND	left-to-right
	Boolean OR	left-to-right

5.2.1 Type Queries

An identifier which has been declared as a type or variable may be queried to get attributes relating to its type. The result is a compile-time constant.

```
min
```

max
size
bits
fbits
align
len – applies only to arrays

An example:

```
type a: _uint8;
var b: 0..31;
var n: _uint;

n = a?size;           // value of 'n' is 1
n = a?bits;           // value of 'n' is 8
n = a?max;            // value of 'n' is 255
n = a?min;            // value of 'n' is 0
n = b?size;           // value of 'n' is 1
n = b?bits;           // value of 'n' is 5
n = b?max;            // value of 'n' is 31
n = b?min;            // value of 'n' is 0
```

5.3 Unary Operations

negation, logical inversion, boolean not

-

~ (one's complement)

! (boolean negation)

5.4 Binary Operations

5.4.1 Multiplicative Operations

multiply, divide, modulo, shifts, and

*

/

%

<<

>>

&

5.4.2 Additive Operations

add, sub, xor, or

+

-

^

|

5.4.3 Comparison Operations

All comparison operations result in the built-in boolean type. Reference types and array types support only equal and not-equal operations. For array types, depending on length and alignment, a call to a library routine may be generated.

==

!=

<

<=

>

>=

5.4.4 Boolean And

&&

5.4.5 Boolean Or

||

5.5 Intrinsic Operations

There are several built-in operations which use a function call syntax.

5.5.1 Absolute Value

The `_abs` operator takes the absolute value of an signed integral number. For example:

```

var a: _int32;
var b: _uint32;

a = -1234;
b = _abs(a);           // value of 'b' is 1123

```

5.5.2 Maximum Value

The `_max` operator takes the largest value of two integral numbers. For example:

```

var a: _int32;
var b: _int32;
var c: _int32;

a = 1234;
b = 5678;
c = _max(a, b);        // value of 'c' is 5678

```

5.5.3 Minimum Value

The `_min` operator takes the smallest value of two integral numbers. For example:

```

var a: _int32;
var b: _int32;
var c: _int32;

a = 1234;
b = 5678;
c = _min(a, b);        // value of 'c' is 1234

```

5.5.4 Square Root

5.5.5 Count Leading Zeros

The `_clz` and `_clznz` operators find the count of the leading (most significant) zeros in an unsigned integer. The `_clznz` operator can be used if the integer is known to be non-zero and may generate more efficient code.

5.5.6 Count Trailing Zeros

The `_ctz` and `_ctznz` operators find the count of the trailing (least significant) zeros in an unsigned integer. The `_ctznz` operator can be used if the integer is known to be non-zero and may generate more efficient code.

5.5.7 Population Count

The `_pop` operator counts the number of 1-bits in an unsigned integer.

5.5.8 Rotate right and rotate left

The `_ror` and `_rol` operators rotate the bits in an unsigned integer. The second argument is the rotation amount and may be an unsigned expression.

5.5.9 Bit Reverse

The `_bitrev` operator reverses the bits in an unsigned integer.

5.5.10 Unpack Byte Array

The `_unpkbe` and `_unpkle` operators take a pointer to an array of bytes and unpacks them into an unsigned integer. The may be done big-endian or little-endian as specified in the last two letters in the operator name.

```
var b: [8]_byte;
var x: _uint16;
var y: _uint32;
var z: _uint64;
x = _unpkbe(b[2:2]); // unpack b[2]..b[3] into 16 bits, big endian
y = _unpkle(b[1:3]); // unpack b[1]..b[3] into 24 bits, little endian
z = _unpkbe(b);      // unpack b[0]..b[7] into 64 bits, big endian
```

5.5.11 Pack Byte Array

The `_packbe` and `_packle` operators take a pointer to an array of bytes and an unsigned integer and pack the integer into the array of bytes. The may be done big-endian or little-endian as specified in the last two letters in the operator name. The result of the pack operators cannot be assigned, so this is more of a statement than an expression.

```
var b: [8]_byte;
var x: _uint16;
var y: _uint32;
var z: _uint64;
_packbe(b[2:2], x); // pack b[2]..b[3] from 16 bits, big endian
_packle(b[1:3], y); // pack b[1]..b[3] from 24 bits, little endian
_packbe(b, z);      // pack b[0]..b[7] from 64 bits, big endian
```

5.5.12 Get Low or High Half

The `_lo` and `_hi` operators return the low and high half, respectively, of an unsigned integer. This currently is implemented for “natural” sized operands.

5.5.13 Create Operand from Two Halves

The `_splice` operator creates an operand from the high and low halves. This currently is implemented for “natural” sized operands.

5.5.14 Set

The `_set` operator initializes an array of byte to a value. The result of the `_zero` operator cannot be assigned, so this is more of a statement than an expression.

```
var array: [32] _byte;
_set(array, ' ');
_set(array[3:8], '#');
```

5.5.15 Zero

The `_zero` operator initializes an object to all zeros. The result of the `_zero` operator cannot be assigned, so this is more of a statement than an expression.

```
var record: {a: _uint64, b: [4]_byte, c: boolean; };
_zero(record);
var array: [32]_uint32;
_zero(array);
```

5.5.16 Find Non-Zero Length

The `_zlen` operator counts the number of non-zero bytes in an array, starting at the beginning. This is useful to determine the length of a string, not including the terminating NUL.

5.5.17 New Allocation

The `_new` operator allocates memory for a new instantiation of a type. For types with an unknown array length, the length of the “unknown” part must be specified in parentheses after the type name. An optional second parameter allows for an implementation specific option (e.g., choice of address space). A library call will be generated for each instance of `_new`.

Examples:

```
type enum: (A, B, C);
type ident:
{   length: _uint;
    name:   []_byte;
};
var pe: @enum;
var pr1, pr2: @ident;
pe = _new(enum);
pr1 = _new(ident(16));           // name array has length 16
pr2 = _new(ident(8), 1);        // use of optional parameter
```

5.5.18 Delete Allocation

The `_delete` operator de-allocates memory for a previously allocated type. As with `_new`, an “unknown” length must be explicitly specified. This allows for memory allocation algorithms which do not store the size of allocated memory in meta-data. Again, as with `_new`, there is an optional second parameter.

Examples, based on the examples in the previous section:

```
_delete(pe);  
_delete(pr1(16));  
_delete(pr2(8), 1);
```

6 Statements

All statements, except for assignment statements, begin with a keyword. All statements end with a semicolon. The declarative statements have been covered previously.

Statement groups start with a “{” and end with a “}”. There is no semicolon after a statement group.

6.1 Declarative Statements

6.1.1 Type Statement

The `type` statement associates a name with type. Refer the the section on types for examples.

6.1.2 Const Statement

The `const` statement creates a named constant. Constants of array and record types may be specified.

```
type rec: { a: _uint8, b: @[]_byte; };  
const rec1: rec = { 42, “foo” };  
type arr: [3]rec =  
{ { 1, “one” }, { 2, “two” }, { 3, “three” } };
```

A `const` statement initialized by a string may trim the terminating NUL by explicit typing.

```
const foo1 = “abcd”;           // foo1?len = 5  
const foo2: [4]_byte = “abcd”; // foo2?len = 4
```

6.1.3 Variable Statement

The `var` statement instantiates a variable. Inside a procedure, the lifetime of the instantiation is from procedure entry to exit. Elsewhere, the lifetime of the variable is the life of the program. Multiple variables with the same type may be created in a single statement.

```
var x, y, z: _uint;
```

6.1.4 Alias Statement

The alias statement can be used for two distinct purposes. The first, more common, use is simply to introduce a new name into the current namespace which is a short-cut to another name. This is usually used to simplify access to names in a package.

The other use of an alias statement is to fix up a forward reference made in a package outside of that

package (perhaps in a subsequent package).

```
package foo
{ type R:
  { next: @R;          // self referential
    what: @P;          // forward reference
  };
}

package bar
{ alias foo.R as R;    // simplify
  type P:
  { next: @P;          // self referential
    from: @R;          // points to R in package foo
  };
  // fixup the forward reference in package foo
  alias P as foo.P;
}
```

6.2 Assignment Statement

6.2.1 Scalar Assignment

The most common example of scalar assignment is when a variable, this case `a`, is assigned a value that is a constant or the value of another variable.

```
a = 0;
a = b;
```

ESL also allows the assignment of multiple variables in a single statement.

```
a, b = 0, 0;
c, d = a, b;
```

In this example, the first statement assigns the variables `a` and `b` a value of zero, while the second statement assigns the values of `c` and `d` with the values of `a` and `b`. After the execution of the second statement the values of `c` and `d` would be zero, respectively.

As will be discussed in more detail in Section 6: Procedures, a procedure can return multiple values. The following example shows how the variables `a` and `b` are assigned the values that are returned from the procedure `sumdiff`.

```
a, b = sumdiff();
```

A more complex example shows a combination of assignments from variables and from the return values of the procedure `sumdiff`.

```
b, x, y, a = a, sumdiff(a, b), b;
```

In all the examples in this section, we have seen one or more variables assigned values with a corresponding number of variables on the right hand side of the assignment statement. Either from a variable or procedure or a combination of the two. In all cases, there has been a one-for-one assignment of a variable from another variable or from the return of a procedure. ESL does not allow for the splitting of a variable into composite types for assignment. In other words, two 16-bit variables cannot be assigned a value from a 32-bit variable on the right hand side. Conversely, a 32-bit variable cannot be assigned the values of two 16-bit variables on the right hand side.

6.2.2 Assignment Operators

Several binary operators can be combined with the assignment operation. They are addition, subtraction, binary or, binary exclusive or, and binary and. The left hand side of the assignment is also the left hand side of the binary operation. For example:

```
x += 1;           // x = x + 1
x -= y+1;         // x = x - (y+1);
x |= y;           // x = x | y;
x[i] ^= y[i];     // x[i] = x[i] ^ y[i];
x.a &= 3;         // x.a = x.a & 3;
```

6.2.3 Record Copy

Records may be copied by simple assignment. The right-hand-side of the assignment may be a record constant. If a record type is “extended” (see section 3.6) it can not be assigned to the base record.

```
var r, s: {a: _uint8; b: boolean; };
s = { 42, true };
r = s;
```

6.2.4 Array Copy

Arrays may be copied by assignment, in whole or in part (i.e. a “slice”). The right-hand-side of the assignment may be an array constant. If an array constant is smaller than the left-hand-side, it will be extended with the last value specified.

There are two assignment operators for array copies. Use of the simple “=” should not be used if the right-hand-side and the left-hand-side may overlap. Instead the assignment operator “=/” should be used.

```
var a, b: [8]_uint8;
a = {42};           // a[0]..a[7] set to 42
b = {1, 2, 3, 4, 9}; // b[4]..b[7] set to 9
b[3:3] = a[0:3];     // b[3]..b[5] set to a[0]..a[2]
a[i:n] = b[0:n];     // a[i]..a[i+n-1] set to b[0]..b[n-1]
a[3:5] =/ a[0:5];    // arrays overlap!
```

6.3 Control Statements

6.3.1 If Statement

The if statement can be a simple if-then with optional else, or a statement which selects among multiple options.

The follow are examples of simple if statements:

```
if foo == 0 then bar = 0;

if bar == 0 then
    foo = 1;
else
{   foo = 2;
    bar = 0;
}

if bar < 10 then foo = 0;
elif bar < 50 then foo = 1;
elif bar < 100 then foo = 2;
else foo = 9;
```

The use of an if statement to select among non-boolean choices is demonstated below:

```
if foo
is 0,2,4..8 then bar = 0;           // if 0,2,4,5,6,7,8
is 1,3,9 then bar = 1;
is 10 then bar = 2;
else bar = 9;
```

6.3.2 Exit Statement

Exit statements must be enclosed in either a loop statement, while statement or do statement. When the boolean expression evaluates to true, the loop is exited. Additional code may optionally be specified to be executed as the exit is taken. See the example in the next section.

6.3.3 Loop Statement

The loop statement is the most general form of loop control. For loops with simple tests at the top or bottom, the while or do statement may be used. The only way to terminate a loop statement is the successful execution of an exit statement.

An example:

```

this = first;
found = false;
loop
{ exit this == 0;
  exit this.value == 0 with found = true;
  this = this.next;
}

```

6.3.4 While/Do Statement

This form of loop statement has the test at the top.

```

while n != 0 do
{
  ...
  n -= 1;
}

```

6.3.5 Do/While Statement

This form of loop statement has the test at the bottom.

```

do
{
  ...
  n += 1;
} while n < 100;

```

6.3.6 For Statement

There are two forms of for statement loops. One is an “in order” sequence, and the other is “order doesn’t matter”. Examples:

```

for k from 0 to n-1 do // loop done sequentially, in order
{
  ...
}

for j in 0..n-1 do      // loop can be done in any order, or in parallel
{
  ...
}

```

6.3.7 Return Statement

The return statement causes an immediate return from a procedure. It will have a list of values corresponding to the return values of the procedure.

```

return x+y, true;

```

6.3.8 Assert Statement

The assert statement allows the programmer to indicate a boolean expression that should always be true at runtime. If asserts are “armed”, and the expression is false, a system-dependent action is taken to

indicate a program error. Assertion arming is disabled by default and may be enabled by a compiler flag. An example of an assertion is:

```
assert n > 0 && n < 100;
```

6.4 Formatting Statements

There are two formatting statements `format` and `print`.

```
format string[(size)],format[,arglist]
print  bufdesc,format[,arglist]
```

To use the `format` statement the package **fmt** must be imported. The `string` is a array of bytes. If processing the format runs out of room, the current contents are removed and processing re-starts of the beginning of the string. If, at compile time, the length of the string is unknown, its size must be explicitly specified. When the format is complete, the string is NUL terminated.

To use the `print` statement the package **prt** must be imported. The `bufdesc` is defined in the package `prt`. The results are put in a buffered output file.

Format is a string that consists of characters which will be directly output and `fmt`-specs. `Fmt`-specs are enclosed in square brackets '[' and ']'. To output a '[', use two '['. `Fmt`-specs are described below.

The `arglist` is a comma separated list of expressions. The format is examined at compile time and the number of format-specs must match the number of expressions in the `arglist`. Also the type of the expression must be compatible with the `fmt`-spec type. This is also checked at compile time.

<i>fmt-spec</i>	=	"[" <i>specs</i>] type "	"
<i>specs</i>	=	[[<i>fill</i>] <i>align</i>][<i>sign</i>][<i>prefix</i>][<i>zero</i>][<i>width</i>][<i>"."</i> <i>prec</i>]	
<i>fill</i>	=	character	(defaults to space)
<i>align</i>	=	"<" ">" "^"	(align left, right, or center)
<i>sign</i>	=	"+"	(force sign even when positive)
<i>prefix</i>	=	"!"	(use prefix such as "0x")
<i>zero</i>	=	"0"	(pad with leading zeros after prefix or sign)
<i>digits</i>	=	decimal number	(0 .. 255)
<i>prec</i>	=	decimal number	(0 .. 255)
<i>type</i>	=	"b"	(binary, prefix is "0b")
		"o"	(octal, prefix is "0o")
		"x"	(hex, lowercase, prefix is "0x")
		"X"	(hex, uppercase, prefix is "0X")
		"p"	(pointer, hex)
		"d"	(decimal)
		"t"	(boolean)
		"f"	(floating point)
		"e"	(floating point, scientific notation)
		"%" <i>ident</i>	(custom format)
<i>ident</i>	=	alpha[alphanumeric*]	

If *width* is not specified, the resulting width will be the minimum necessary to output the item.

Formats can be extended with custom formats which may be different for each program. The *ident* will be suffixed with "fmt". See the section on Custom Formatting.

Example of format:

```
import fmt;
proc foo(buf: @[]_byte, n: _uint)
{
    format buf(512), "centered prefix |[!16X]|", n;
}
```

The results in buf are: "centered prefix | 0xF123 |".

Example of print statement:

```
import prt;
proc foo(bd: fmt.pDesc, s: @[]_byte, j: _int)
{
    print bd,    "[*>16s] [+8d]\n", s, j;
}
proc main(): _int
{
    var bd: fmt.pDesc;
    bd = prt.open(prt.fdo, 256); // stdout, bufsiz=256
    foo(bd, "string", 1234);
    prt.close(bd);              // flush and close
    return 0;
}
```

The above example outputs the line: "*****string +1234"

6.5 Asm Statement

The asm statement is used to insert an assembly language instruction.

The first string is the assembly language prototype. The prototype may contain replacement variables indicated by a '\$' followed by a digit. The second (optional) string is the constraint list. The statement ends with a list of expressions to be used as arguments. An example:

```
asm "sumdiff $0 $1 $2 $3", "=r,=r,r,r", ret1, ret2, arg1, arg2;
```

7 Procedures

Procedures may return zero or more values. Returned values can be expressed in one of two syntax forms:

- with a colon followed by a list of types
: _uint, _boolean
- between parentheses as a list of names with types
(foo: _uint, ok: _boolean)

When returning multiple values, the first form may be deprecated in the future.

7.1 Normal Procedures

Normal procedures are as expected. In the following example, the procedure ‘sumdif’ is defined to have two parameters and return two values. In the first syntax form this is:

```
proc sumdiff(a:_int, b:_int):_int, _int
{
    return a+b, a-b;
}

var x, y: _int;
...
x, y = sumdiff(x, y);
```

In the second syntax for, the procedure sumdiff becomes:

```
proc sumdiff(a:_int, b:_int)(sum: _int, diff: _int)
{
    sum, diff = a+b, a-b;
}
```

Note that a return statement is not required, Any of the returned values that are not explicitly assigned are set to zero.

7.2 Methods

Methods allow the programmer to write a procedure associated with a particular type. (This is similar to “methods” in other languages, but the binding is to a type not a “class”.) In the following example, the type ‘foo’ is declared and a method that operates on that type is defined:

```
type foo: 0..100;

proc (a: foo) saturate(b: _uint): foo
{
    var sum: _uint;
    sum = a + b;
    if sum > foo?max then sum = foo?max;
    return foo(sum);
}

kernel-devel-5.16.11-100.fc34.x86_64
var x: foo;
...
x = x.saturate(10);
```

While the previous example showed a method to a scalar type, a more usual case is when the type is a record:


```

type rec: { a:_uint; b:_boolean };

proc (p:@rec) init(x:_uint)
{
    p.a = x;
    p.b = true;
}

var r: rec;
...
r.init(0);

```

7.3 Procedure Attributes

7.3.1 Inline/Noinline

Inline and noinline is an attribute only of a procedure, i.e., proc statement. The inline attribute is currently only a hint to the compiler to tell it to “try very hard to inline”. The noinline attribute tells the compiler to never inline the procedure. Procedures without the inline or noinline attribute may be inlined, unless a compiler flag is used to disable inlining globally.

7.3.2 Global

The global attribute for procedures is used just as it is for variables.

7.3.3 Weak

The weak attribute for procedures is used just as it is for variables.

7.3.4 External

The external attribute for procedures is used just as it is for variables.

```

package foo
{
    proc foobar(): : external;
};

```

When the external attribute is used with a procedure, the name that will be generated in the symbol table will be the unqualified name. In other words, the name will be that as it appears in the proc statement without the preface of the package name that it may be in.

External with a parameter for a procedure.

```

package cpu
{
    proc getcpuid(result: @[]_uint32): : external(0x1fff1ff1);
};

```

Here the code for 'getcpuid' is known to be located at address 0x1fff1ff1.

7.3.5 Segment

The section attribute for procedures is used in a similar manner as it is for variable, see section. 4.2.4. The format of section names is linker-specific.

Example:

```
proc add(a: _int): _int: section(".simple_functions")
{ ... }
```

7.4 Parameter Attributes

The formal parameters to a procedure may have a single attribute: **in**, **out**, or **io**. These are used when the parameter is by reference to indicate whether the reference object is read-only, write-only or both. Currently this is not enforced and therefore is for readability only.

7.5 Forward Procedures

In some cases, a procedure must be called before it is defined (as with mutually recursive procedures). A procedure may be declared forward of its use but before its definition by omitting the body of the procedure and replacing it with a single semicolon:

```
proc sumdiff(a:_int, b:_int)(sum:_int, diff:_int);
```

7.6 Procedure References

A procedure reference is a data type which contains a pointer to a procedure with a given "signature". The signature is the parameter list together with the returned value list. This is prefixed by '@_' to indicate this is a pointer to an unnamed procedure.

```
type pref: @_(a:_int, b:_int)(sum:_int, diff:_int);
var p: pref;
p = sumdiff;
```

References to methods have the same syntax as those to normal procedures. The first parameter is that of the method parameter.

```
type bpref: @_(a:@rec, x:_uint);
var bp: bpref;
p = r.init;
```

8 Packages

Packages are the primary method of reusing code by providing a private namespace. A package is a bundle of declarations and procedures wrapped in a package name.

Examples:

```
package foo:
{ type type1: 0..1000;
  type type2: (no, yes, maybe);
  proc sub1(x: type1, y:type 1): type2
  { statement-list }
}
```

8.1 Package Continuation

Packages can be re-opened to extend the contents of the package. This is known as *package continuation*. An example is:

```
package foo
{
  var x: _uint32;
  var y: _uint32;
}

// code outside package foo

package foo          // continue package foo
{
  var z: _uint32;
}                    // package foo now has variable x,y,z
```

A package can be imported into another package. This is not *package continuation*, but rather the definition of a new package containing a nested package.

Supposed a file named "pkg_A.esl" contains:

```
package pkg_A
{
  var v1: _uint32;
  var v2: _uint32;
};
```

In a different file, is the following code:

```

package pkg_B
{
    import pkg_A;

    var v1: _uint32;
    var v2: _uint16;
    proc x()
    {
        v1 = 33;
        v2 = 44;
        pkg_A.v1 = 55;
        pkg_B.v2 = 66;
    }
};

```

9 Programs and Scope

Programs consist of global declarations, packages, procedures and import statements.

9.1 Import Statement

The import statement causes file inclusion. The included file typically contains one or more package, but this is not a requirement. There are several variations of import statements:

```

import foo;                // imports the file foo.esl
import "y/foo" "bar";     // imports the file y.foobar.esl
import (<expr>);          // imports the result of a string expression

```

A file can be imported only once per namespace. Subsequent imports in the same namespace are ignored.

Package specifiers can be long. When a package is imported, the package name can be renamed with the `alias` statement. This enables short names to be used in the program.

Alternatively, individual identifiers in the package can be aliased. In that way, the package name does not have to be used as a prefix on every reference to an identifier within that package.

9.2 Conditional Compilation

If statements may be used outside of procedures to implement conditional compilation. It is suggested that full bracketing with “{” and “}” be used. All code in the if statement must have legal syntax.

9.3 Compilation Style

Most programmers have adopted an incremental compile approach to program development. In other words, one compiles numerous source files, one at a time, and then links the resultant object files together to form a single, executable file. In the ESL / LLVM ecosystem, this approach is not necessary nor desirable.

Program development with ESL is best done with a 'compile all at once' approach. Simply put, all the files associated with a particular development are compiled at once. This approach to compilation gives the ESL front-end, and to a greater extent the LLVM back-end, a complete view of all the software in a development project. By using this approach it allows ESL and the LLVM tools to maximize the optimization of the source code in the development project. There can be great advantages to code size and execution efficiency:

1. Procedures (non-global) that are not called do not generate any code.
2. Variables and constants (non-global) not used are not allocated.
3. Opportunities to in-line procedures are maximized.
4. Procedures called with an actual argument that is always the same are simplified.

The following shows how five files in a development are combined and compiled all at once, using ESL. The main file in this development is: `top_level.esl`. It imports the four other files in the development into `top_level.esl`.

=== `top_level.esl`

```
import file1;
import file2;
import file3;
import file4;

proc main(argc: _uint, argv: @[]@[] _byte): _int
{
    .
    .
    .
    .
}
```

10 Custom Formatting

All programs that use the `print` or `format` statement must import the package `fmt`. To implement a custom format, the `fmt` package is extended. The following is an example that formats an IPv4 address into a dotted decimal string. In a program, the `print` statement might be:

```
var ipaddr = { 1, 200, 32, 0 };
```

```
print bd, "[*^20%ipv4]\n", ipv4addr; // pad '*' centered with 20 byte field
```

The output line would be:

```
"*****1.200.32.0*****\n"
```

The extended `fmt` package might look like this:

```

import prt;                // also imports fmt

const IPv4AddrLen = 4;
type IPv4Addr: [IPv4AddrLen]_byte: align(4);

package fmt                // extend the fmt package with new stuff
{
    // The routines decwid and deccvt are pre-defined in fmt.
    // The method padding is also pre-defined and returns
    // where the payload should be stored, the width of the
    // payload, and the width of any zero prefix (not used here).

    proc ipv4cvt(buf: _address, a: @IPv4Addr): _uint
    {
        var i, j, k: _uint;
        k = 0;
        i = 0;
        loop
        {
            j = decwid(a[i]);    // width without leading zeros
            deccvt(buf[k:], j, a[i]);    // convert decimal
            k += j;
            exit i == IPv4AddrLen-1;
            buf[k] = '.';
            k += 1;
            i += 1;
        }
        return k;
    }
}

// The following method is called when executing the format.
// The conversion is done in a temporary buffer because we
// don't know the width ahead of time.

proc (bd: pDesc) ipv4fmt(a: @IPv4Addr, spec: FmtSpec)
{
    var w: _uint;
    var tmp: [15]_byte;
    var buf: @_memory;
    w = ipv4cvt(tmp, a);
    buf, w, _ = bd.padding(w, RIGHT, spec);
    buf[0:w] = tmp[0:w];
}
}

```

11 Language Syntax Summary

11.1 Notation

The syntax is specified in EBNF (Extended Backus-Naur Form). Lexical symbols are enclosed in double quotes.

11.2 Program Syntax

```
program      = prog-stmt { prog-stmt }
prog-stmt    = package | procedure |
              import-stmt | decl-stmt | if-stmt | error-stmt
import-stmt  = "import" package-specifier ";"
error-stmt   = "error" string ";"
```

11.3 Package Syntax

```
package      = "package" ident "{" pkg-stmt { pkg-stmt } "}"
pkg-stmt     = package | procedure |
              import-stmt | decl-stmt | if-stmt | error-stmt
```

11.4 Procedure Syntax

```
procedure    = "proc" [ "(" formal ")" ] ident signature [ ":" proc-attr ]
              ( ";" | "{" { stmt } "}" )
signature    = "(" [ parm-list ] ")" [ ":" [ retv-list ] ]
parm-list    = formal { "," formal }
formal       = ident ":" type-def
retv-list    = type-def { , type-def }
stmt         = "{" stmt-list "}" |
              decl-stmt |
              asgn-stmt | if-stmt | loop-stmt | while-stmt | exit-stmt | return-stmt | assert-stmt |
              asm-stmt
```

11.5 Declaration Statement Syntax

decl-stmt = *alias-stmt* | *type-stmt* | *var-stmt* | *const-stmt*
alias-stmt = "alias" *alias-list* ";"
alias-list = *alias-spec* { ", " *alias-spec* }
alias-spec = *qname* "as" (*ident* | *pkg-name* "." *name*)
type-stmt = "type" *ident* ["(" *typeid* ")"] ":" *type-def* [":" *type-attr-list*] ";"
var-stmt = "var" *ident* { ", " *ident* } ":" *type-def* [":" *var-attr-list*] ";"
const-stmt = "const" *ident* [":" *type-def*] "=" *constant* ";"
type-def = *type-name* | *type-range* | *type-enum* | *type-ref* | *type-record* | *type-array*
type-range = *number* "." *number*
type-enum = "(" *enum-list* ")"
enum-list = *enum-def* { ", " *enum-def* }
enum-def = *ident* ["=" *sconst*]
type-ref = "@" *type-def*
type-record = "{" *field* { *field* } "}"
field = *ident* ":" *type-def* [":" *field-attr-list*] ";"
type-array = "[" *type-index* "]" *type-def*
type-name = *name*
type-attr-list = *type-attr* { ", " *type-attr* }
type-attr = "bits" "(" *cexpr* ")" | "size" "(" *cexpr* ")" | "align" "(" *cexpr* ")" |
"lsb" | "msb" |
"le" | "be" |
"ro" | "wo" | "at"

11.6 Executable Statement Syntax

asgn-stmt = *lhs-list* "=" *expr-list* ";"
 lhs "+=" *expr* ";" | *lhs* "-=" *expr* ";"
 lhs "|=" *expr* ";" | *lhs* "&=" *expr* ";" | *lhs* "^=" *expr* ";"
if-stmt = "if" *bool-expr* "then" *stmt* ["else" *stmt* |
 "if" *expr* *is-list* ["else" *stmt*]
is-list = *is-clause* [*is-list*]
is-clause = "is" *is-value-list* "then" *stmt*
is-value-list = *is-value* { ", " *is-value-list* }
is-value = *cexpr* ["." *cexpr*]
loop-stmt = "loop" *stmt*
while-stmt = "while" *bool-expr* "do" *stmt*
exit-stmt = "exit" *bool-expr* ["then" *stmt*]
return-stmt = "return" *expr-list* ";"
assert-stmt = "assert" *bool-expr* ";"
asm-stmt = "asm" *string* [", " *string* { ", " *expr* }] ";"

11.7 Expression Syntax Summary


```

cexpr      = expr # compile time constant
expr       = baexpr { " | " baexpr }
baexpr     = cmpexpr { "&&" cmpexpr }
cmpexpr    = addexpr [ cmpop addexpr ]
cmpop      = "==" | "!=" | "<" | ">" | "<=" | ">="
addexpr    = mulexpr { addop addexpr }
addop      = "+" | "-" | "|" | "^"
mulexpr    = uexpr { mulop mulexpr }
mulop      = "*" | "/" | "%" | "<<" | ">>" | "&"
uexpr      = term | unop term
unop       = "+" | "-" | "~" | "!"
term       = type-query | number | qname | "(" expr ")"
type-query = [ type-name | var-name ] "?" type-qattr
type-qattr = "min" | "max" | "bits" | "size" | "align" | "len"

```

12 Library Support

```

memcpy
memset
memcmpN
memlenN
memalloc
memfree
_assert

```

13 ESL For C Programmers

Since C has been the dominant system programming language for the past 30 years or so, this chapter is designed to help C programmers to understand how to express C idioms as ESL idioms. For those programmers who have been exposed to "Pascal-like" languages (e.g. Module, Ada, etc.), some of this might be familiar.

One of the major differences between C and ESL is that ESL has no reserved words! You are free to name a variable "if" or a type "then" or a procedure "else".

13.1 Preprocessor

ESL does not have a separate pre-processor language. There are no features such as "token-pasteing."

13.1.1 Include

The ESL "package" and "import" features replace the C "header files" and "#include" mechanism.

13.1.2 If and ifdef

There is no equivalent to "#ifdef". However the "#if" pre-processor feature has an ESL equivalent. The if statement can be used if its selection expression is a compile-time constant. Be aware that the ESL if statement deals in units of complete statements, not tokens, when doing conditional compilation.

13.2 Declarations

A major difference between C and ESL is that ESL uses the Pascal-like syntax for declarations. In ESL the identifier comes first followed by the type. In C, the reverse is true.

13.3 Typedefs

The ESL type statement is the equivalent of the C typedef.

13.4 Enumerations

The identifiers representing values in a C enumeration are exposed in the same name space as the enumeration. In ESL (as in some other languages) the identifier needs to be exposed by giving its type name as a prefix. However, in many cases, the type can be inferred and the enumeration prefix is not necessary.

13.5 Pointers and Arrays

C example:

```
typedef int *pi;           // pointer to int
typedef int ai[8];         // array of int
typedef int *api[8];       // array of pointers to int
typedef int (*pai)[8];     // pointer to array of int
```

ESL example:

```
type pi: @_int;           // pointer to int
type ai: [8]_int;         // array of int
type api: [8]@_int;       // array of pointers to int
type pai: @[8]_int;       // pointer to array of int
```

13.6 Pointers to Procedures

In C, the declaration of a pointer to a function has a syntax that has confused even experienced programmers. Consider a pointer to a function that takes two integer arguments and returns an integer:

```
typedef int (*func)(int, int);
```

In ESL, the syntax is still a little funky due to the use of the null identifier “_”, but perhaps easier to read:

```
type func: @(a:_int, b:_int): _int;
```

Or the example from page 122 of the 2nd edition of Kernighan and Ritchie, x is defined as an “array[3] of pointer to function returning pointer to array[5] of char”:

```
typedef char ((*x[3])())[5]);
```

Which in ESL is written pretty much as the text describes it:

```
type x: [3]@_(): @[5]_byte;
```

13.7 Parameters and Arrays

In C function parameters are passed “by value” *except for arrays*. In the case of an array, the address of the array is passed instead. In ESL all procedure parameters are passed “by value”, with no exceptions. If you want a pointer to an array, then the procedure parameter must be so declared.

Also, in C, pointer arithmetic may be used as a mechanism to access elements in arrays. This is not allowed in ESL, access to the element must be done with array indexing.

For example, the Unix command line convention passes an a pointer to an array of pointers to strings as an argument vector (e.g. “argv”). In C this is often specified as:

```
int main(int argc, char *argv[])
{
}
```

In ESL all the initial pointers and the specification of a string as “[]_byte” must be explicit:

```
proc main(argc: _int, argv: @[ ]@[ ]_byte): _int
{
}
```

13.8 Statements

13.8.1 Assignment Statements

Multiple assignment is not allowed in ESL. C statements such as

```
x = y = 0;
```

must be split into multiple assignment statements in ESL as in the following:

```
x = 0;
y = 0;
```

Alternatively, as mentioned in page 26, the following could be used:

```
x, y = 0, 0;
```

13.8.2 Assignment Within an Expression

In ESL, expressions cannot have side effects (except for procedure calls), such C constructs must be written as a separate assignment statement.

13.8.3 If Statements

If statements in ESL are very similar to those in C. The syntax is a little different, the parentheses surrounding the expression are not necessary, but the **then** keyword is.

13.8.4 Switch Statements

ESL doesn't have an explicit statement type corresponding to the C “switch” statement. The ESL “if” statement covers the same territory with one important difference - ESL doesn't allow “fall-through”. Also each ESL “case” is a single statement, if more than one is required than a statement group delimited by “{” and “}” is required.

C Example:

```
unsigned int foo;
switch (foo) {
    case 0: case 1: case 7: case 8: case 9: case 15:
        statement1;
        break;
    case 2: case 3: case 4:
        statement2;
        statement3;
    case 5:      /* fall thru */
        statement4;
        break;
    default:
        statement5;
        break;
}
```

ESL Example:

```
var foo: 0..15;
if foo
is 0..1, 7..9, 15 then
    statement1;
is 2..4 then
{ statement2;
  statement3;
  statement4; // no fall thru
}
is 5 then
    statement4;
else
    statement5;
```

14 Building and Using the ESL Compiler

14.1 Prerequisites

Before one builds the ESL compiler, two packages must be install: the LLVM Compiler Infrastructure and the GNU Compiler Collection (only the assembler and libc are used) for the target platform. The building and/or installation of both of these packages are described on their respective web pages and will not be repeated here.

14.2 Building the ESL compiler

14.2.1 Getting the source

After the LLVM and GNU collections have been installed, the source to ESL can be downloaded from Github:

```
git clone https://github.com/bagel99/esl
```

For the experimental fixed point support:

```
cd esl
git checkout master
```

14.2.2 Installing LLVM

14.2.3 Building the compiler

Once the sources to ESL have been obtained, the next step is to actually build and install the compiler. The top level directory of the distribution should contain two files: LICENSE and README, and two directories: src and doc. The building and installing of the compiler will occur in the src directory and all of the following directions will assume that the user has that directory as the current working directory.

Before starting the build process, a line must be changed in the shell script file: llvm. The line that set the shell variable LLVM must be changed to the directory where the LLVM tools, i.e., llc, llvm-as, etc., were installed into. The line looks like the following:

```
LLVM=$HOME/work/llvm/Release+Asserts/bin          # !!FIX THIS!!
```

Once this line has been changed, building can proceed.

After the line in LLVM has been changed, the make(1) variables: CFLAGS, ARCH, and GCC should be examined and possibly adjusted for the target platform. The file: Makefile is the makefile used to build the ESL compiler and contains the definitions of the aforementioned variables. Once these two files have been examined and optionally changed, the ESL compiler can be built.

To build the ESL compiler, the user invokes 'make'. The make'ing of the ESL will proceed and produce four (4) versions of the compile: eslc0, eslc1, eslc2, and eslc3. The user should select the 'eslc3' version as the ESL compiler to be used for compilation. This binary can be put into the user's choice of a directory and making sure the directory defined in the user's shell path variable.

14.3 ESL Command Line Options

```
eslc [-D[asf]] [-mtarget] [-Iidir] [-ofile] [-Ooopt] [-Aaopt] [-M] [-g] files
```

14.3.1 -D[asf]

Control debug output:

- a - Dump the abstract syntax tree
- s - Dump the symbol table
- f - List all source files

14.3.2 -mtarget

Specify the target architecture.

ESL can only support what LLVM has implemented as target backends.

Current targets include:

x86

x86_64

x86_64-darwin

msp430

cortex-m3

arm920t

ppc32

ppc64

s390x

systemz

mips

my66000

14.3.3 -Idir

Add a directory to the set of directories search for import files. They are searched in the order specified.

14.3.4 -ofile

14.3.5 -Ooopt

s - compile for small size

z - compile for minimum size

14.3.6 -Aaopt

a – arm assertions

i– tag all procedures as no-inline

z – zero local scalar variables

14.3.7 -Ffopt

t – warn on truncation

a – warn on ambiguous statement

r – warn on old-style return value syntax

14.3.8 -M

Instead of compiling, just output a list of files suitable for makefile dependencies.

14.3.9 -g

Generate debugging information for use by run-time debuggers such as gdb. [This doesn't work yet.]

14.4 Compilation Flow

eslc – the compiler “front-end”

opt – LLVM architecture independent optimization

llc – LLVM “back-end” that generates assembly language [Future versions of LLVM may generate object code and eliminate the need for a target assembler.]

gas – target assembler

ld – linker used to link in library routine

Alphabetical Index